

## Discussion-5 Benefits of using programming standards and guidelines

### **Discussion Topic:**

Please choose one of the following questions to discuss in your initial post:

- What are the main benefits of using programming standards and guidelines?
- What are some of the challenges of applying programming standards and guidelines in a team environment?
- How can we balance the need for flexibility and innovation with the need for consistency and maintainability in software development?.

### **My Post:**

Hello Class,

For this discussion, I chose the “What are the main benefits of using programming standards and guidelines?” question topic.

Using programming standards and guidelines comes with many benefits.

Programming standards play an essential role in software development (BrowserStack, 2024).

- They help maintain consistency in code development when working alone or within a team of programmers (CSU Global, 2025).
- They reduce the workload for integrators, maintainers, and testers by eliminating the need to deal with diverse software algorithms and implementations.
- They simplify the summarization of code in documentation.
- They enable the use of automated tools to validate standardized software and identify dependencies.
- They facilitate the use of design patterns with implementations.
- They promote Low coupling, high cohesion, well-defined interfaces, flexibility, and product evolution in stages

In other words, programming standards help teams create high-quality code and software, that is, they ensure that software is safe, secure, and reliable for the developers and the users (Codacy Platform, 2025)

Programming guidelines are coding best practices that can be defined as a set of methods that can be applied to implement programming standards (CSU Global, 2025).

- They make code easier to read and to understand, they improve the readability of the code.
- They promote modularity, that is, the implementation of applications from independent, reusable blocks
- They promote code generalization, that is, the creation of adaptable code.

- They promote dependency visibility using relationships and coupling between components through the use of parameter names and comments.
- They guide the design of algorithms by considering performance, even though this increases efficiency, may involve drawbacks in development efforts (AlgoCademy, n.d.).

In other words, programming guidelines are coding best practices that help implement programming standards by promoting writing efficient, reliable, and maintainable code (Patel, 2024). They help to address common issues and pitfalls in programming.

-Alex

### References:

AlgoCademy (n.d.). The missing piece: *Why you need to learn algorithm design before writing code*. AlgoCademy. <https://algotcademy.com/blog/the-missing-piece-why-you-need-to-learn-algorithm-design-before-writing-code/>

BrowserStack (2024, June 28). *Coding standards and best practices to follow*. BrowserStack. <https://www.browserstack.com/guide/coding-standards-best-practices>

Codacy Platform (2025, March 11). *Coding standards: What are they and why are they important?* Codacy. <https://blog.codacy.com/coding-standards>

CSU Global (2025). *Module 5: Project Implementation* [Interactive Lecture]. Canvas. [https://csuglobal.instructure.com/courses/110425/pages/module-5-overview?module\\_item\\_id=5733321](https://csuglobal.instructure.com/courses/110425/pages/module-5-overview?module_item_id=5733321)

Patel, R. (2024, June 5). *Understanding coding standards: Why they matter*. medium. <https://medium.com/@ravipatel.it/understanding-coding-standards-why-they-matter-cc75adcba245>